SALES ANALYTICS DASHBOARD

Power BI | BI Portfolio Case Study

Power BI • **DAX** • **Data Modeling** • **SQL** • **Power Query** • **Stakeholder Reporting**

£2.04M Total revenue analyzed	**625** Orders / transactions	**13** Months of data	**4** Dashboard pages
7 Products tracked	**5** Regions compared	**64%** On-time delivery rate uncovered	**94%** Revenue volatility flagged (West)

Project	Sales Analytics Dashboard — Power BI Portfolio Case Study
Role	BI Analyst / Power BI Developer (solo project)
Timeframe	13-month transactional dataset (Dec 2023 – Dec 2024)
Tools	Power BI Desktop, DAX, Power Query, SQL, Excel
Deliverables	Interactive 4-page Power BI dashboard, PBIX file, video walkthrough

Project Summary

A self-directed Power BI project simulating a real-world sales analytics engagement: 13 months of order-level transactional data across 5 regions and 7 products, transformed into a four-page interactive dashboard that serves executives, regional managers, and operations leads from a single data model. The project moves beyond “what happened” reporting into diagnostic analysis — explaining why revenue moved, where risk is concentrated, and what a stakeholder should act on next.

The Business Problem

Sales, regional, and operations teams typically work from separate reports: revenue summaries from finance, fulfillment metrics from operations, and product performance from category managers — each refreshed on a different cadence, in a different format, using inconsistent definitions of “growth” or “on-time.” That fragmentation creates three concrete problems this project was built to solve:

- No single source of truth — stakeholders reconcile numbers across reports before a conversation can even start.
- Averages hide risk — topline metrics (e.g. average delivery time) can look healthy while a meaningful share of transactions are failing underneath them.
- Growth and volatility get confused — without a consistent, rule-based classification, a region having one large swing quarter can be read as “growing” when it's actually unstable.

Project Goal

Design and build a single Power BI solution that gives every stakeholder — from an executive scanning for 30 seconds to a regional manager doing a deep dive — a consistent, filterable, drill-through-capable view of sales performance, product mix, regional trends, and delivery reliability, built on dynamic DAX logic that stays correct as new months of data are added.

My Approach

- Modeled the data as a star schema — a central SalesData fact table joined to Product_Dim and a region dimension — keeping filters fast and calculations consistent across all four report pages.
- Replaced a traditional Calendar-table dependency with YearMonth string-based navigation using EDATE, after diagnosing that the Calendar-table approach was breaking Rolling 3-Month Average and MoM variance logic under certain filter contexts.
- Authored a full library of DAX measures: rolling 3-month revenue, MoM growth %, dynamic growth-category classification (High Growth / Moderate / Decline), and a dynamic three-tier color-coding measure using SWITCH(TRUE()) and RANKX for the product revenue ranking chart.
- Designed each page around a single stakeholder question rather than a generic KPI wall — Executive Summary answers “is the business healthy,” Product & Region answers “where should we focus,” Operations answers “what's the delivery risk.”
- Validated findings against the raw order-level drill-through table before writing any insight callout, to make sure every number in a callout box could be traced back to source rows.

The Solution — Dashboard Structure

Four linked pages, sharing one filter panel (Year / Region / Product), so any slicer selection updates every page consistently.

1. Executive Summary

Purpose: *30-second health check for leadership.*

- Total revenue, order count, average order value, and top region at a glance
- 13-month monthly revenue trend with 3-month rolling average overlay
- Growth-category donut (High Growth / Moderate / Decline / No Prior Period)
- Top 5 products by revenue with an auto-generated executive insight narrative

2. Sales Performance

Purpose: *Trend and variance analysis for sales leadership.*

- YTD revenue, rolling 3-month revenue, and MoM revenue change KPI cards
- Combo chart: monthly revenue bars against a MoM growth % line
- Monthly stacked area chart showing regional revenue composition shift over time
- Product × Region revenue matrix with a white-to-navy heat scale for fast pattern-spotting

3. Product & Region

Purpose: *Where to focus resources next quarter.*

- Full product revenue ranking, sorted descending
- Revenue-vs-volume scatter plot (bubble size = average unit price) to separate “high price, low volume” products from “volume champions”

- Revenue by region and quarter, grouped column chart
- Auto-written Product Intelligence and Region Intelligence callouts summarizing the strategic read on each

4. Operations & Trends

Purpose: *Fulfillment reliability and risk detection.*

- Average / max delivery days and on-time delivery rate against a 7-day target
- Delivery speed distribution (Fast / Normal / Slow bands)
- Delivery performance by region against a 5-day target line
- Growth category distribution by month (100% stacked) and a drill-through order detail table

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Key Insights

- **Revenue reached £2,041,646 across 625 orders over 13 months, with October 2024 the peak month at £192,662, driven by Electronics demand in the North region.**
- A mid-year slowdown emerged in Aug–Sep 2024, with revenue dropping to roughly £130K/month — invisible on a YTD total but clear on the monthly trend.
- **West is the most volatile region — a 94% revenue swing between April and June 2024 — which reads as instability, not growth, once isolated from North's steady lead.**
- Phone is the weakest product overall (£192K revenue, 75 orders) and hits its lowest point specifically in the Central region (£14K) — a strategic-review candidate rather than a marketing-spend problem.
- **On-time delivery sits at 64% (400 of 625 orders ≤ 7 days) despite an average delivery time of 6.2 days that looks comfortably within target — a textbook case of an average masking a distribution problem.**
- Furniture (Chair + Desk) represents 25% of total revenue — a meaningful share that doesn't show up in the top-line product ranking, where both sit mid-table individually.

Technical Highlights (DAX)

A selection of the more advanced logic built into the model — useful for a technical interview walkthrough:

- Dynamic Three-Tier Colour Measure — SWITCH(TRUE()) combined with RANKX to color-code the product revenue ranking chart by relative position, wrapped in explicit CALCULATE-based context transition so ranking evaluates correctly per product rather than breaking under visual-level filter context.
- Rolling 3-Month Average — rebuilt after diagnosing that a Calendar-table-anchored version was producing incorrect variance and “previous month” values; resolved by switching to YearMonth string navigation with EDATE, removing the dependency that was causing the errors.
- Dynamic growth-threshold classification — High Growth / Moderate / Decline bands are calculated relative to MoM % change rather than hardcoded, so the classification stays accurate as new months are appended without manual rework.

Business Recommendations Delivered

The purpose of the dashboard isn't just visualization — it's decision support. Three recommendations this analysis surfaces for a stakeholder audience:

- Investigate Phone's underperformance in Central specifically — the single weakest product×region combination in the matrix, and a clearer, more actionable signal than “Phone is weak” in isolation.
- Treat West's volatility as an operations/fulfillment question first, not a demand-generation one — a 94% swing over three months needs root-causing before it's read as a trend either direction.
- Escalate the on-time delivery rate (64% against a 7-day target) ahead of the average delivery time metric — it's the number most likely to be quietly eroding repeat business.

Skills Demonstrated

Technical

- Power BI report design and data modeling (star schema)
- DAX — CALCULATE, RANKX, SWITCH(TRUE()), EDATE, time-intelligence and rolling-window measures
- Power Query for data shaping and transformation
- SQL for source data extraction and validation
- Root-cause debugging of filter-context and evaluation-context errors in DAX

Analytical & Business

- KPI design for multiple stakeholder audiences from a single model
- Diagnostic analysis — explaining drivers of change, not just reporting totals
- Risk detection — identifying where averages obscure distribution problems
- Business storytelling — translating dashboard output into stakeholder-ready recommendations

What This Project Demonstrates

This case study is designed to show hiring managers evidence of the full BI analyst workflow — not just the ability to build charts: data modeling decisions made and justified, DAX bugs diagnosed and fixed (not just written once and left), and a finished product that leads with business judgment (“here's what to do about it”) rather than description (“here's what the numbers are”).

Links & Downloads

- Interactive dashboard walkthrough (video): [Insert YouTube link]
- Power BI file (.pbix): [Insert download / GitHub link]
- Full DAX documentation: [Insert link, if published separately]
- Contact: [Insert LinkedIn / email]